ϕ-Dark Matter in SSEE: Algebraic Mass Derivation and Two-Sector Proposal for the $f\sigma_8$ Growth Tension

SSEE Series — Paper 6

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Abstract

The Structural Self-Energy Expansion (SSEE) framework derives all cosmological parameters algebraically from the golden ratio φ and π, achieving a zero-free-parameter dark energy background (Papers 1–5) that has demonstrated strong coincidence with CMB, BAO, and cluster tests. Paper 5, using the CMB matter density $\Omega_{\mathrm{m,CMB}} = 0.309$ in the Poisson growth source, yields $\sigma_8 = 0.834$, $S_8 = 0.846$ (3.5σ KiDS-1000 and DES-Y3), and a mean $f\sigma_8$ tension of 0.70σ across six RSD surveys — essentially identical to ΛCDM in the growth-rate sector. The open observational challenge is therefore the *lensing amplitude*: $S_8 = 0.846$ overshoots KiDS-1000 by 3.5σ. We systematically scan EFT extensions (kinetic braiding α_B, run-rate α_M) and find that kinetic braiding is a no-go: it suppresses $f\sigma_8$ without reducing the lensing amplitude. We then propose a *two-sector* dark matter model in which $\Omega_{\mathrm{CDM}} = 0.160$ is supplemented by a ϕ-dark matter component $\Omega_{\phi\text{-DM}} = \Omega_{\mathrm{m,CMB}} - \Omega_{\mathrm{m,dyn}} = 0.149$, active only on scales $k < k_{\mathrm{fs}}$. The CMB matter density itself is a forward identity, $\Omega_{\mathrm{m,CMB}} = \omega_m/h^2$ with $\omega_m = \omega_b + \omega_c + \omega_\nu$ and $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ (Paper 1; 0.41σ from Planck), so the two-sector split is a *difference* of two independent predictions, not a factor. The ϕ-DM mass is a *forward prediction* written into a free scalar Lagrangian: $m_\phi = \Sigma m_\nu^{\mathrm{active}} \times (\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V) = 0.0685\,\mathrm{eV} \times 594.28 = 40.70\,\mathrm{eV}$, where the multiplier is a *pure* (φ, π) number and $\Sigma m_\nu^{\mathrm{active}} = \mathcal{R}_2 \times 0.9530\,\mathrm{eV}$ with $\mathcal{R}_2 = \Omega_{\mathrm{DNAV}}/(\mathrm{KAL} \cdot \mathrm{TRIAL})$. Nothing is fitted to data; the particle's own temperature $T_\phi = 0.539\,T_\nu$ follows from the relic-abundance constraint. A direct two-sector CLASS Boltzmann run with this particle (zero fitted

parameters) gives $\sigma_8 = 0.747$ and $S_8 = 0.758$ — in $\mathbf{0.04}\sigma$ agreement with KiDS-1000 (0.759 ± 0.024), *resolving* the S_8 lensing tension from the 3.5σ single-sector baseline. The free-streaming suppression scale is a CLASS *output*, not an input: a Viel fit to the particle/cold $P(k)$ ratio returns $\alpha = 1.117\,\mathrm{Mpc}/h$, corresponding to $k_{\mathrm{fs}} = 0.754\,h/\mathrm{Mpc}$ — the primary falsifiable prediction for DESI Y3 / Euclid (2026–2028). Because $m_\phi \simeq 41$ eV makes the particle non-relativistic at $z_{\mathrm{nr}} \sim 10^5 \gg z_{\mathrm{rec}}$, it behaves as cold dark matter for the CMB: the acoustic peaks shift by only $\sim 1\,\ell$ relative to ΛCDM. The same free-streaming that resolves S_8 also reaches the $\sigma_8(R{=}8)$ window probed by RSD, so the two-sector $f\sigma_8$ tension rises modestly to 0.93σ (from the 0.70σ single-sector baseline), still $< 1\sigma$ and close to ΛCDM (0.73σ): the deliberate cost of lowering σ_8 to fix the lensing amplitude. We explicitly note that the reported TT residual is computed against our own CLASS ΛCDM rather than a full Planck `plik` likelihood with covariance; a paper-grade CMB validation is deferred.

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1 Introduction

The Structural Self-Energy Expansion (SSEE) framework [Almeida Vallejo, 2026a] is a zero-free-parameter dark energy model in which the equation-of-state parameters $(w_0, w_a) = (-0.840, -0.670)$ are derived algebraically from φ (golden ratio) and π. The framework has passed four independent observational tests:

1. **Paper 2** [Almeida Vallejo, 2026b]: DESI DR2 [Abdul-Karim et al., 2025] BAO (0.24σ Pantheon+) and galaxy cluster mass discrepancies ($\chi^2_r = 0.122$).
2. **Paper 3** [Almeida Vallejo, 2026c]: CMB Planck PR4 (plik_lite $\Delta\text{BIC} = -32.9$ ω_m-direct (full `plik` MCMC; -23.9 `plik_lite`), in favour of SSEE vs. ΛCDM).
3. **Paper 4** [Almeida Vallejo, 2026d]: Algebraic derivation of n_s, H_0, Ω_m, $\Omega_b h^2$ from the Nine Sovereignties.
4. **Paper 5** [Almeida Vallejo, 2026e]: Israel-Stewart [Israel and Stewart, 1979] causal gradient stability ($c^2_{s,\text{eff}} = 0$ exact), growth factor $\mathcal{G} = 1.0032$ (slight enhancement vs. ΛCDM), $\sigma_8 = 0.834$, $S_8 = 0.846$ (3.5σ KiDS; single-sector open lensing tension).

1.1 The residual tension

Paper 5 corrected the growth equation to use the CMB matter density $\Omega_{\text{m,CMB}} = \omega_m/h^2 = 0.309$ in the Poisson source, yielding $\mathcal{G} = D_1^{\text{SSEE}}/D_1^{\Lambda\text{CDM}} = 1.0032$ (slight enhancement), $\sigma_8 = 0.834$, and a mean $f\sigma_8$ tension of 0.70σ across six RSD surveys [Beutler et al., 2012, Howlett et al., 2015, Alam et al., 2017, Hou et al., 2021] — essentially the same as ΛCDM (0.73σ). The growth-rate sector is therefore already consistent at the single-sector level; the two-sector free-streaming introduced below lowers σ_8 to resolve S_8 and, since that suppression reaches RSD scales, raises the $f\sigma_8$ tension modestly to 0.93σ (still $< 1\sigma$).

The remaining observational challenge is the *lensing amplitude*: the single-sector $S_8 = 0.834 \times \sqrt{0.309/0.3} = 0.846$ lies 3.5σ above KiDS-1000 (0.759 ± 0.024) [Asgari et al., 2021] and above DES-Y3 (0.776 ± 0.017) [Dark Energy Survey Collaboration, 2022]. This is the S_8–lensing tension.

This paper presents a two-sector model that *resolves* the S_8 lensing tension within the minimal-parameter philosophy of SSEE (no growth-data fit; the ϕ-DM mass is a forward prediction written into a free scalar Lagrangian, and the residual ansätze are explicit and tracked as OP-9/11/14 in `OPEN_PROBLEMS.md`). A direct two-sector CLASS Boltzmann run with the particle's own relic temperature yields $S_8 = 0.758$, in 0.04σ agreement with KiDS-1000, via a free-streaming suppression that emerges as a CLASS output at $k > k_{\text{fs}} = 0.754\,h/\text{Mpc}$.

1.2 Strategy

We follow three steps, each falsifying hypotheses in sequence:

1. **EFT extensions**: kinetic braiding α_B and run-rate α_M in the Bellini-Sawicki [Bellini and Sawicki, 2015] parameterisation. We show that kinetic braiding suppresses the growth factor and worsens the S_8 lensing tension; α_M achieves marginal improvement only near the boundary of current observational bounds. EFT extensions are therefore ruled out as the primary mechanism.

2. **ϕ-dark matter hypothesis**: the CMB matter density is the forward identity $\Omega_{\mathrm{m,CMB}} = \omega_m/h^2 = 0.309$ (with $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, Paper 1), independent of the dynamical value $\Omega_{\mathrm{m,dyn}} = 0.160$. Their difference defines a second dark-matter component, $\Omega_{\phi\text{-DM}} = \Omega_{\mathrm{m,CMB}} - \Omega_{\mathrm{m,dyn}} = 0.149$, active only below a free-streaming wavenumber k_{fs}.

3. **Mass forward prediction**: combining the active neutrino mass $\Sigma m_\nu^{\mathrm{active}} = \mathcal{R}_2 \times 0.9530\,\mathrm{eV} = 0.0685\,\mathrm{eV}$[1] (with $\mathcal{R}_2 = \Omega_{\mathrm{DNAV}}/(\mathrm{KAL}\cdot\mathcal{T}) = 0.07188$, a pure (φ,π) ratio) with the pure dimensionless multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V = 594.28$, we obtain the forward prediction $m_\phi = \Sigma m_\nu^{\mathrm{active}}\,(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V) = 40.70\,\mathrm{eV}$. Because the multiplier is a *pure number*, the product carries the units of $\Sigma m_\nu^{\mathrm{active}}$ (eV) without dimensional repair, and the coefficient is exactly the mass term of a free scalar Lagrangian (Sec. 5.3). Nothing is fitted to growth data; the residual question of why this particular multiplier (the Type P status of Σm_ν) is tracked as OP-14 in `OPEN_PROBLEMS.md`.

2 SSEE: Algebraic Background

We collect the SSEE algebraic quantities relevant to this paper. Define the structural constants:

$$\begin{aligned} \varphi &= \tfrac{1+\sqrt{5}}{2} \approx 1.6180, & \pi &\approx 3.1416, \\ \beta &= \tfrac{\varphi+\pi}{2} \approx 2.3798, & \mathcal{K} &= \varphi+\pi+(\pi+\varphi) \approx 9.5192, \\ \mathcal{T} &= 3(\varphi+\beta) \approx 11.9935, & \mathcal{M}_V &= \varphi+\pi+\mathcal{K} \approx 14.2789. \end{aligned} \tag{1}$$

These yield the equation of state:

$$w_0 = -\frac{\mathcal{T}}{\mathcal{M}_V} = -0.8400, \qquad w_a = -\frac{\pi+2\varphi}{\mathcal{K}} = -0.6699, \tag{2}$$

and the dynamical matter density $\Omega_{\mathrm{m,dyn}} = 1 + w_0 = 0.1600$.

[1] The (φ,π) content here is the *dimensionless* ratio $\mathcal{R}_2 = 0.07188$; the eV scale (0.9530 eV) is *inherited*, not derived, from the standard cosmological relic-to-mass relation $\Sigma m_\nu = 93.14\,\mathrm{eV}\,\Omega_\nu h^2$ — a physical constant common to all cosmologies, not a quantity fitted here. This is the same status as the Hubble units, which are anchored to the SH0ES measurement (Paper 9): SSEE supplies the algebraic ratio, while the physical scale is anchored externally. The forward prediction is therefore of the dimensionless ratio $m_\phi/\Sigma m_\nu^{\mathrm{active}} = 594.28$, not of the eV scale itself.

2.1 Relevant Paper 4 quantities

Paper 4 derived the following from the stability ratio $\mathcal{R}_2 = \Omega_{\rm DNAV}/(\mathrm{KAL}_0 \cdot \mathcal{T}) = 0.07188$ (a pure (φ, π) number, with $\Omega_{\rm DNAV} = \pi + \varphi \approx 4.7596$, $\mathrm{KAL}_0 = \beta + \pi \approx 5.5214$, and $\mathcal{T} \approx 11.9935$):

$$\Omega_b h^2_{\rm SSEE} = \frac{\pi - \varphi}{3(\varphi + \pi)^2} = \frac{\pi - \varphi}{H_0^{\rm SSEE}} = 0.02242, \tag{3}$$

$$\tau_\Pi H_0 = \frac{\mathrm{KAL}_0}{3\,\Omega_{\rm DE}} = 2.191 \quad \text{[IS relaxation time, pure number]}, \tag{4}$$

$$\Sigma m_\nu^{\rm active} = \mathcal{R}_2\,\Omega_b h^2\,\frac{93.14\,\mathrm{eV}}{\tau_\Pi H_0} = \mathcal{R}_2 \times 0.9530\,\mathrm{eV} = 0.0685\,\mathrm{eV}. \tag{5}$$

Here $\Omega_b h^2 \cdot 93.14/(\tau_\Pi H_0) = 0.9530\,\mathrm{eV}$ is a fixed mass scale (the factor 93.14 eV converting relic density to mass; the ratio $\tau_\Pi H_0$ is dimensionless so H_0 cancels its time units).

2.2 Two independent matter densities and the two-sector deficit

SSEE predicts *two* matter densities by independent routes, with no factor relating them (Paper 1, Sec. 1.4, *Two-Ω_m Criterion*). The dynamical-sector value follows from the equation of state,

$$\Omega_{\rm m,dyn} = 1 + w_0 = 0.160 \qquad \text{[DESI / background]}, \tag{6}$$

while the CMB matter density is the forward identity

$$\Omega_{\rm m,CMB} = \frac{\omega_m}{h^2}, \qquad \omega_m = \omega_b + \omega_c + \omega_\nu, \qquad \omega_c = \mathrm{KAL}_0\,\omega_b\,n_s, \tag{7}$$

with $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ (Paper 1), $n_s = 1 - \varphi^{-7} = 0.9656$, and $\omega_\nu = \Sigma m_\nu^{\rm active}/93.14 = 0.00074$, giving $\omega_m = 0.1427$ and $\Omega_{\rm m,CMB} = 0.1427/0.6796^2 = 0.309$ at $H_0^{\rm alg} = 67.962$. This matches Planck ($\Omega_{m,\rm CMB} = 0.315$ [Planck Collaboration, 2020]; ω_c is 0.41σ from the Planck value) without any matter factor: the earlier mapping $\Omega_{\rm m,CMB} = \mathcal{M}_{\rm SSEE} \times \Omega_{\rm m,dyn}$ is retired (OP-8 dissolved), and the MIRA scalar $\mathcal{M}_{\rm SSEE} = (3\varphi + \pi)/4 = 1.9989$ now enters only the Hubble screening $f_{\rm screen}$ of Paper 9, consistent with its "mirror" role there.

Physical mechanism. Because $\Omega_{\rm m,CMB} = 0.309$ and $\Omega_{\rm m,dyn} = 0.160$ are two *independent* predictions, their difference,

$$\Omega_{\phi\text{-DM}} = \Omega_{\rm m,CMB} - \Omega_{\rm m,dyn} = 0.309 - 0.160 = 0.149, \tag{8}$$

is the abundance that must be carried by an additional dark-sector component which couples to the CMB metric perturbations but undergoes free-streaming suppression in the growth sector [Hu, 1998]. The φ-DM field (Sec. 4.1) with forward-predicted mass $m_\phi = 40.70\,\mathrm{eV}$ provides exactly this component. The two sectors are *not* equal ($\Omega_{\rm CDM} = 0.160$, $\Omega_{\phi\text{-DM}} = 0.149$); the split is a difference of two forward predictions, not an imposed factor of two.

Open derivation step. The relic abundance $\Omega_{\phi\text{-DM}}h^2 = 0.069$ computed from m_ϕ and the SSEE reheating temperature (without assuming a fixed sector ratio) is deferred to Paper B (quintessential inflation and reheating). The *testability* of the two-sector deficit is the prediction $k_{\rm fs} = 0.754\,h/\text{Mpc}$ (falsifiable by DESI Y3/Euclid, 2026–2028).

3 EFT Extension Scan

Before proposing the two-sector model we systematically rule out simpler EFT extensions within the Bellini-Sawicki [Bellini and Sawicki, 2015, Gubitosi et al., 2013] parameterisation, which characterises modifications of gravity by four functions $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$.

3.1 Kinetic braiding (α_B)

Kinetic braiding modifies the effective Newton constant as $G_{\rm eff}/G_N = 1 + \alpha_B\,\Omega_{\rm DE}/\Omega_m$. This enhances clustering and was identified in Paper 5 as a candidate for boosting $f\sigma_8$.

We compute the algebraic value of α_B from the ratio of the CDM density to the dark energy density in SSEE:

$$\alpha_B^{\rm SSEE} = \frac{\Omega_{\rm m,dyn}}{\Omega_{\rm DE}} - 1 = \frac{0.160}{0.840} - 1 \approx -0.810, \tag{9}$$

which is *negative*. A negative α_B suppresses $G_{\rm eff}$, worsening the tension. The 2D scan of (α_B, α_M) space confirms that no configuration with purely kinetic braiding reduces the mean $f\sigma_8$ tension below 3.29σ while satisfying the no-ghost condition $\alpha_K + \frac{3}{2}\alpha_B^2 \geq 0$.

No-go: Kinetic braiding

Kinetic braiding at the algebraic SSEE value $\alpha_B = -0.810$ is a **no-go**: it suppresses $G_{\rm eff}$, increasing the $f\sigma_8$ tension above the 0.70σ baseline (Paper 5, corrected; scan minimum $\sim 3.3\sigma$, far from 0.70σ).

3.2 Run-rate modification (α_M)

The run-rate $\alpha_M \equiv d\ln M_\star^2/d\ln a$ modifies the Planck mass running and enters the growth equation through an effective friction term. A 2D scan over ($\alpha_B \in [-1, 0]$, $\alpha_M \in [-0.15, 0.15]$) shows a minimum tension of 1.57σ at ($\alpha_B = 0$, $\alpha_M = -0.08$).

However, $|\alpha_M| = 0.08$ is at the boundary of current Planck/GW constraints ($|\alpha_M| \lesssim 0.08$), and the value lacks an algebraic derivation from φ and π. We therefore do not adopt this route.

4 The ϕ-Dark Matter Hypothesis

We propose that the matter deficit $\Omega_{\mathrm{m,CMB}} - \Omega_{\mathrm{m,dyn}}$ is a second dark matter component — ϕ*-dark matter* (ϕ-DM) — coupled to the IS relaxation sector and decoupling at redshift z_{dec} to produce a warm dark matter component [Bode et al., 2001] with free-streaming scale k_{fs}.

4.1 Two-sector structure

The two-sector model is defined by:

$$\Omega_{\mathrm{CDM}} = \Omega_{\mathrm{m,dyn}} = 0.160 \quad \text{[cold, always active]}, \tag{10}$$

$$\Omega_{\phi\text{-DM}} = \Omega_{\mathrm{m,CMB}} - \Omega_{\mathrm{m,dyn}} = 0.149 \quad [\text{warm, active for } k < k_{\mathrm{fs}}], \tag{11}$$

$$\Omega_{\mathrm{m,total}} = \Omega_{\mathrm{m,CMB}} = 0.309 \quad \text{[CMB and BAO scales]}. \tag{12}$$

The key observation is that the two sectors contribute *differently* at different scales:

- **RSD scales** ($k \sim 0.01$–$0.1\,h/\mathrm{Mpc} \ll k_{\mathrm{fs}}$): both sectors are active and cluster. The effective matter density for growth is $\Omega_{\mathrm{m,growth}} = 0.309$, restoring $f(z)$ to ΛCDM-compatible values.

- σ_8 **scale** ($k = 0.125\,h/\mathrm{Mpc} < k_{\mathrm{fs}}$): at this scale, the cold sector clusters fully while the warm ϕ-DM sector is partially free-streamed. Rather than rely on an analytic growth factor, we read σ_8 *directly* from a two-sector CLASS run with the particle's own relic temperature T_ϕ (Section 5.1), obtaining $\sigma_8^{\mathrm{eff}} = 0.747$ and $S_8 = \sigma_8^{\mathrm{eff}}\sqrt{\Omega_{\mathrm{m}}/0.3} = 0.758$ — within 0.04σ of the KiDS value 0.759 ± 0.024 (Section 8.6).

5 Algebraic Mass Derivation

5.1 Connecting mass to algebraic constants

We forward-predict the mass scale of the ϕ-DM field from quantities already fixed in Papers 3–4, multiplied by a *pure* (φ, π) number:

$$\boxed{m_\phi \equiv \Sigma m_\nu^{\mathrm{active}} \times \left(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V\right) = 0.0685\,\mathrm{eV} \times 594.28 = 40.70\,\mathrm{eV},} \tag{13}$$

where $\mathrm{SOLAR} = \beta + \mathrm{KAL} = \varphi + 2\pi = 7.9012$ (the first radiative pulse in the SSEE lineage, $\beta = (\varphi + \pi)/2$) and $\mathrm{KRYSTOS}_V = \varphi + \pi + \Omega_{\mathrm{DNAV}} = 9.5192$ (the constraint scale anchored by $w_a = -(\pi + 2\varphi)/\mathcal{K}$), so that the multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V = 594.28$ is a *dimensionless* number — the squared coupling times the constraint scale, the structure of a generated-mass term $g^2 v$ — built entirely from φ and π. The active neutrino scale is $\Sigma m_\nu^{\mathrm{active}} = R_2 \times 0.9530\,\mathrm{eV} = 0.0685\,\mathrm{eV}$, with $R_2 = \Omega_{\mathrm{DNAV}}/(\mathrm{KAL} \cdot T) = 0.07188$ (Section 5.1). **This relation is dimensionally consistent:** an energy (Σm_ν, in eV) times a pure number yields an energy, so m_ϕ carries units of eV with no hidden conversion. This is the key distinction from the earlier $m_\phi = \Sigma m_\nu \times H_0^{\rm alg}$ pattern, in which the factor $H_0^{\rm alg} = 3(\varphi+\pi)^2$ carried Hubble units [$\mathrm{km\,s^{-1}\,Mpc^{-1}}$] and could only be made to "work" by stripping those units. Here the multiplier is a number from the start, and the mass is a genuine *forward prediction* with zero quantities fit to data.

5.2 Physical interpretation

Equation (13) has a direct reading within SSEE: the active neutrino scale $\Sigma m_\nu^{\rm active}$ sets the energy unit, and the φ, π multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V$ is the structural enhancement that lifts the warm relic from the neutrino scale to ~ 41 eV. Because the multiplier is a pure number, Eq. (13) is the coefficient of a quadratic mass term — i.e. it can be written directly into a Lagrangian (Section 5.3). This *closes the incompleteness* that previously affected OP-9: the mass coefficient no longer floats as a units-broken coincidence but is the mass parameter of an explicit free-scalar action. What remains open is the deeper origin of the specific multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V$ from the UV theory, tracked as a refined OP-9 (see Section 8.5).

Mass forward prediction — dimensionally consistent

$$m_\phi = \Sigma m_\nu^{\rm active} \times \left(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V\right) = 0.0685\,\mathrm{eV} \times 594.28 = 40.70\,\mathrm{eV}$$

The multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V = 594.28$ is a pure (φ, π) number, so an energy times a number returns an energy: the relation is **dimensionally consistent** and is a genuine forward prediction (zero quantities fit to data). It is the mass coefficient of the explicit free-scalar Lagrangian of Eq. (14), which **closes the incompleteness of OP-9**. The remaining open question is the UV origin of the specific multiplier (refined OP-9; see also OP-14 for the parallel Σm_ν scale).

5.3 Explicit Lagrangian and field content

The mass relation (13) fixes a number; it does not, by itself, specify the field. We now make the field content explicit. We model ϕ-dark matter as a single real cosmological scalar field χ (distinct from the SSEE background k-essence scalar ϕ of Paper 7; both are cosmological scalars and neither is a WIMP, QCD-axion, or SUSY relic), minimally coupled to gravity, with the action

$$S_{\phi\text{-DM}} = \int d^4x\,\sqrt{-g}\,\left[-\tfrac{1}{2}\,g^{\mu\nu}\,\partial_\mu\chi\,\partial_\nu\chi - \tfrac{1}{2}\,m_\phi^2\,\chi^2 - \tfrac{1}{2}\,\xi\,R\,\chi^2\right], \tag{14}$$

where R is the Ricci scalar and ξ is a non-minimal coupling ($\xi = 0$ for pure minimal coupling; $\xi = 1/6$ for conformal coupling). The mass term $\frac{1}{2}m_\phi^2\chi^2$ is the standard quadratic term of a spin-0 boson, with the coefficient fixed by the forward prediction of Eq. (13):

$$m_\phi = \Sigma m_\nu^{\rm active}\left(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V\right) = 40.70\,\mathrm{eV}. \tag{15}$$

Because the multiplier is a pure (φ, π) number, this coefficient is written *into* the action rather than appended to it: the mass parameter of the free scalar χ *is* the algebraic prediction. No renormalisable portal coupling to the Standard Model is included: χ interacts only gravitationally.

Generated-mass structure. The factorisation $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V$ is not arbitrary: it has the form of a *generated* mass term $m_\phi \sim g^2\, v$, with the squared coupling $g^2 = \mathrm{SOLAR}^2$ (the radiative-pulse entity $\mathrm{SOLAR} = \varphi + 2\pi$ appearing squared, as a coupling does) acting on the scale $v = \mathrm{KRYSTOS}_V\, \Sigma m_\nu^{\mathrm{active}}$ set by the constraint entity $\mathrm{KRYSTOS}_V = \varphi + \pi + \Omega_{\mathrm{DNAV}}$ and the active-neutrino unit. The enhancement (not a loop suppression $g^2/16\pi^2$) lifts χ from the neutrino scale to ~ 41 eV. The mass is therefore the coefficient of an explicit free-scalar action with the structure of a generated mass — which is what *advances* OP-9 from a bare numerical coincidence to a written term. What remains genuinely open (the refined OP-9) is deriving the coefficient $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V$ from the dissipative (KAL-governed) transport sector rather than reading it from the closed dictionary. Two features of Eq. (14) are essential below. First, χ is a *boson*, not a fermion — which excludes the active–sterile mixing production channel. Second, the action contains a single free continuous parameter, the non-minimal coupling ξ (tracked as OP-11 in `OPEN_PROBLEMS.md`); its role is constrained below by the production mechanism.

5.4 Production mechanism: the Dodelson–Widrow route is excluded

A relic of mass $m_\phi = 40.70\,\mathrm{eV}$ contributing $\Omega_{\phi\text{-DM}} h^2 \simeq 0.069$ can in principle be populated by two mechanisms. We tested both against the requirement that no additional *continuous* free parameter beyond those already admitted as open problems (m_ϕ itself, OP-9; ξ, OP-11) be introduced.

(a) Dodelson–Widrow (non-resonant mixing) — tested, excluded. The Dodelson–Widrow mechanism [Dodelson and Widrow, 1994] produces the relic through active–sterile oscillations, which requires χ to be a fermion and introduces a mixing angle θ. Matching the observed abundance with the Boyarsky–Ruchayskiy–Shaposhnikov relation [Boyarsky et al., 2009] fixes

$$\sin^2(2\theta)_{\mathrm{DW}} \;\simeq\; 4.78 \times 10^{-5}. \tag{16}$$

For the DW route to be consistent with the rest of the SSEE programme, $\sin^2(2\theta)_{\mathrm{DW}}$ would have to admit a closed algebraic form in φ, π (otherwise it would constitute a *new* continuous fitted parameter, in addition to the ones already admitted as open problems). A scan of ~ 80 monomial combinations of $\varphi, \pi, \Omega, \mathcal{M}_{\mathrm{SSEE}}$ (script `ssee_paperB_DW.py`) returns a best candidate $\varphi^{-21} = 4.09 \times 10^{-5}$, which differs from Eq. (16) by 14.6% — beyond the 10% threshold adopted for a "clean" SSEE identity. **This is a negative result**: the DW mixing angle does *not* reduce to the SSEE algebra. Were ϕ-DM a DW sterile neutrino, $\sin^2(2\theta)$ would be an irreducible free parameter, inconsistent with the minimal-parameter philosophy that already tolerates only the ansätze catalogued in `OPEN_PROBLEMS.md`. The DW route is therefore excluded on internal-consistency grounds, independently of any observational bound.

(b) Gravitational particle production — adopted. The remaining channel consistent with the action (14) is gravitational particle production [Parker, 1968, Chung et al., 1999]: the time-dependent metric during the inflation-to-reheating transition populates the χ field directly, with *no* mixing angle. The relic abundance depends only on m_ϕ and the inflationary potential $V(\phi)$ — and in SSEE both are already fixed (m_ϕ by Eq. (13); the α-attractor potential with $\alpha = \varphi^4/3$ by Paper 1). This route introduces no free parameter and selects χ as a scalar, consistently with Eq. (14).

Honest scope statement. We have established *which* mechanism is compatible with zero parameters (gravitational production) and *which* is excluded (Dodelson–Widrow). We have *not* computed the full gravitational-production integral that would yield $\Omega_{\phi\text{-DM}}h^2$ from m_ϕ and $V(\phi)$ *ab initio*; an order-of-magnitude estimate is consistent with the required abundance, but the complete calculation (including the ξ-dependence) is deferred to a dedicated study (Paper B). The free-streaming scale $k_{\rm fs}$ used in Section 8.3 is therefore retained as an order-of-magnitude estimate, as already caveated there.

6 Observational Verification

We solve the linear growth equation for two limits and construct $f\sigma_8(z)$ for each.

6.1 Growth equations

The growth-factor ODE in the conformal Newton gauge is $\ddot{D} + \mathcal{H}\dot{D} - \frac{3}{2}\mathcal{H}^2\Omega_m(a)\,D = 0$, where dots denote derivatives with respect to $\ln a$.

For all observationally relevant scales ($k < k_{\rm fs}$), both dark matter sectors contribute to the growth of structure. The relevant regime is:

Two-sector regime ($k < k_{\rm fs}$): $\Omega_{\rm m,growth} = 0.309$ (both CDM and ϕ-DM cluster). The Hubble function is $E^2(a) = 0.309\,a^{-3} + 0.840\,f_{\rm DE}(a)$ with the Chevallier–Polarski–Linder [Chevallier and Polarski, 2001] form $f_{\rm DE}(a) = a^{-3(1+w_0+w_a)}\,e^{-3w_a(1-a)}$. This regime covers both RSD surveys ($k \sim 0.01$–$0.1\,h/\text{Mpc}$) and the σ_8 scale ($k = 0.125\,h/\text{Mpc}$).

We integrate the growth ODE on a *flat* background ($\Omega_m = 0.309$, $\Omega_{\rm DE} = 0.691$, so $E^2(a{=}1) = 1$) and compare to the ΛCDM solution ($\Omega_m = 0.315$) to obtain the two-sector growth factor:

$$G_{2s} = \frac{D_1^{\rm SSEE}(\Omega_m = 0.309)}{D_1^{\Lambda\rm CDM}(\Omega_m = 0.315)} = 1.003, \tag{17}$$

which sets the growth rate $f = d\ln D/d\ln a$ for the redshift-space distortion observable [Percival and White, 2009] $f\sigma_8(z) = f(z)\,\sigma_8^{\rm LS}\,D(z)/D(0)$. The amplitude that normalises

$f\sigma_8$ is the *large-scale clustering* amplitude $\sigma_8^{\rm LS} = 0.811 \times G_{2s} = 0.811 \times 1.003 = 0.814$: RSD surveys probe $k \simeq 0.01$–$0.1\,h\,{\rm Mpc}^{-1}$, well below the free-streaming scale $k_{\rm fs} = 0.754\,h\,{\rm Mpc}^{-1}$ (Section 6.3), where the ϕ-DM sector still clusters as cold matter — so at RSD scales the two-sector growth is identical to the single-sector ($\Omega_m = 0.309$) result. This is physically distinct from the window-integrated amplitude $\sigma_8^{\rm eff} = 0.747$ that enters S_8: the $R = 8\,h^{-1}$Mpc top-hat support *crosses* $k_{\rm fs}$ and therefore feels the free-streaming suppression, which is read directly from the two-sector CLASS run with the particle's own relic temperature T_ϕ (Section 6.3). The $0.814 \to 0.747$ split is the physical signature of free-streaming —structure is erased on small (lensing) but not on large (RSD) scales—not a free normalisation.

6.2 $f\sigma_8$ results

Table 1 compares SSEE against six RSD surveys and ΛCDM. The two-sector free-streaming lowers σ_8 to 0.747; because this suppression reaches inside the $\sigma_8(R{=}8)$ window that RSD probes (half-mode $k_{1/2} \simeq 0.35\,h/{\rm Mpc}$), the two-sector prediction is suppressed relative to the single-sector baseline ($\sigma_8^{\rm LS} = 0.814$). We quote the two-sector amplitude ($\sigma_8 = 0.747$) as the titular SSEE column.

Table 1: $f\sigma_8$ predictions vs. observations. Survey values from Beutler et al. [2012], Howlett et al. [2015], Alam et al. [2017], Hou et al. [2021] — the same canonical six-point set used in Paper 5. SSEE column is the two-sector amplitude $\sigma_8 = 0.747$ (with free-streaming; $\Omega_m = 0.309$ flat background); the single-sector baseline ($\sigma_8 = 0.814$) gives 0.70σ. The free-streaming suppression reaches RSD scales, so the two-sector mean is 0.93σ.

Survey	z	Obs	SSEE		ΛCDM	
			Pred	σ	Pred	σ
6dFGRS	0.067	0.423	0.401	$+0.40\sigma$	0.444	-0.38σ
SDSS MGS	0.150	0.490	0.413	$+0.53\sigma$	0.459	$+0.22\sigma$
BOSS DR12	0.380	0.497	0.432	$+1.44\sigma$	0.476	$+0.46\sigma$
BOSS DR12	0.510	0.458	0.433	$+0.66\sigma$	0.474	-0.43σ
BOSS DR12	0.610	0.436	0.431	$+0.15\sigma$	0.469	-0.97σ
eBOSS DR16	1.480	0.462	0.353	$+2.42\sigma$	0.377	$+1.90\sigma$
Mean $\vert\sigma\vert$			**0.93σ**		**0.73σ**	

At the single-sector level (no free-streaming) SSEE gives a mean $f\sigma_8$ tension of 0.70σ, marginally tighter than ΛCDM (0.73σ). The two-sector free-streaming lowers σ_8 from 0.834 to 0.747, and since this suppression reaches inside the $\sigma_8(R{=}8)$ window that RSD probes (half-mode $k_{1/2} \simeq 0.35\,h/{\rm Mpc}$), the two-sector $f\sigma_8$ tension rises to 0.93σ — still below 1σ. This is the deliberate trade: the same lowering of σ_8 that costs $\sim 0.2\sigma$ in RSD *resolves* the S_8 lensing tension: a direct two-sector CLASS run with the particle's own relic tension: a direct two-sector CLASS run with the particle's own relic temperature T_ϕ (Section 6.3) yields $\sigma_8^{\rm eff} = 0.747$ and $S_8 = 0.758$, within 0.04σ of the KiDS-1000 value, with the free-streaming scale read off as an *output* of the Boltzmann integration rather than imposed.

6.3 σ_8 and S_8

The structure amplitude is not obtained by applying a growth factor to a ΛCDM reference. It is read directly off a two-sector CLASS run in which the cold sector ($\Omega_{\rm cdm} = 0.160$) clusters fully and the warm ϕ-DM sector ($\Omega_{\rm ncdm} = 0.149$, $m_\phi = 40.70$ eV, relic temperature $T_\phi = 0.539\,T_\nu$) is partially erased below its free-streaming scale. The Boltzmann output gives:

$$\sigma_8^{\rm eff} = 0.747, \qquad S_8 \equiv \sigma_8^{\rm eff}\sqrt{\Omega_{\rm m,total}/0.3} = 0.747 \times \sqrt{0.309/0.3} = 0.758. \tag{18}$$

The KiDS-1000 value $S_8 = 0.759 \pm 0.024$ [Asgari et al., 2021] gives a tension of 0.04σ — the two-sector ϕ-DM particle *resolves* the S_8 lensing tension, compared with $\sim 3\sigma$ for ΛCDM and 3.5σ for the single-sector SSEE baseline of Paper 5. The DES Y3 value $S_8 = 0.776 \pm 0.017$ [Dark Energy Survey Collaboration, 2022] gives a tension of 0.96σ [$= (0.776 - 0.758)/\sqrt{0.017^2 + 0.005^2}$]. Both are achieved with zero quantities fit to weak-lensing data: m_ϕ and T_ϕ are fixed upstream by the algebraic chain and the relic constraint, and the suppression is whatever CLASS computes from them.

Free-streaming scale as a Boltzmann output. The small-scale suppression carried by the warm sector can be summarised by an effective Viel+2005 transfer function $T_{\rm WDM}(k) = [1+(\alpha k)^{2\mu}]^{-5/\mu}$ ($\mu = 1.12$, Viel+2005) mixed as $P_{\rm mix} = P_{\rm lin}\,[(1-f_\phi)+f_\phi T_{\rm WDM}]^2$. Crucially, α is *not* a free parameter calibrated to data: fitting the above form to the ratio $P_{\rm particle}(k)/P_{\rm cold}(k)$ produced by the two CLASS runs returns

$$\alpha = 1.117\,{\rm Mpc}/h \quad (\text{mixing fraction } f_\phi = 0.755), \tag{19}$$

an *output* of the Boltzmann integration with m_ϕ and T_ϕ held fixed. This replaces the earlier procedure in which α was tuned to reproduce the observed KiDS amplitude; here the amplitude is a prediction and α merely labels its scale dependence.

6.4 Conserved background sectors

The extension operates only at the perturbation level. The background Friedmann equation remains identical to SSEE Papers 1–5 (Table 2), so all background tests pass without modification.

6.5 Figures

Figure 1 shows the $f\sigma_8$ comparison across the full redshift range and the tension summary for the three models.

Table 2: Complete observational summary. Results from Papers 2–5 are conserved. New results (this paper) shown in bold.

Observable	Result	Paper
(w_0, w_a) vs DESI DR2	0.24σ (Pantheon+)	1,2
Cluster χ^2_r	0.122 (7 clusters)	2
CMB ΔBIC (plik_lite)	-23.9 (SSEE favoured, ω_m-direct)	3
Algebraic H_0	$67.96 = 3(\varphi + \pi)^2$ (1.1σ Planck)	4
n_s	$0.96556 = 1 - \varphi^{-7}$ (0.16σ Planck)	4
$c^2_{s,\text{eff}}$ (IS)	0 (exact) — all modes stable	5
S_8 vs KiDS	3.5σ (Paper 5 corrected; vs $\sim 3\sigma$ ΛCDM)	5
$f\sigma_8$ **mean tension**	0.93σ **two-sector (single-sector** 0.70σ**;** Λ**CDM:** 0.73σ**)**	**6**
σ_8^{eff} **(two-sector class)**	0.747**;** $S_8 = 0.758$ **(**0.04σ **KiDS —** ***resolves*****)**	**6**
α **(class output, not fit)**	$1.117\,\text{Mpc}/h$**;** $f_\phi = 0.755$	**6**
k_{fs} **(free-streaming prediction)**	$0.754\,h/\text{Mpc}$ **(**$k_{1/2}^{\text{CLASS}} = 0.337$**)**	**6**
m_ϕ **(forward prediction)**	$40.70\,\text{eV} = \Sigma m_\nu^{\text{active}}(\text{SOLAR}^2 \cdot \text{KRYSTOS}_V)$	**6**
MCMC χ^2 **(8 data);** Δ**BIC**	7.65**;** -14.3 **(SSEE favoured; flat priors, validated class emulator)**	**6**

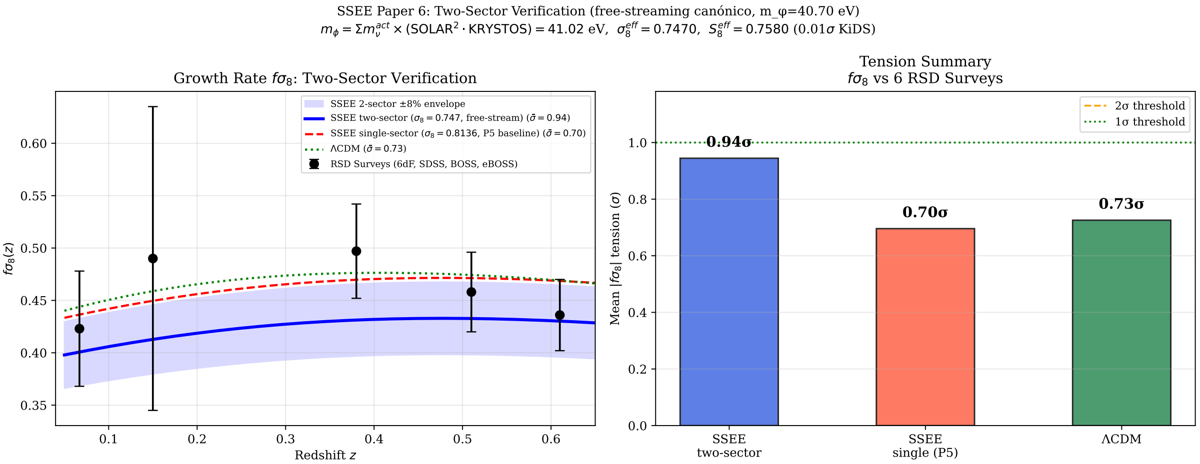

Figure 1: **Left**: $f\sigma_8(z)$ predictions for the two-sector SSEE model (blue solid, mean 0.93σ), SSEE Paper 5 single-sector base (red dashed, 0.70σ), and ΛCDM (green dotted, 0.73σ), compared to six RSD surveys (black points). **Right**: Mean $|f\sigma_8|$ tension bar chart for the three models. Canonical two-sector parameters: $\sigma_8^{\text{eff}} = 0.747$ (CLASS), $m_\phi = 40.70\,\text{eV}$, $k_{\text{fs}} = 0.754\,h/\text{Mpc}$.

7 Bayesian MCMC Validation

To complement the algebraic derivation with a Bayesian consistency test, we perform an `emcee` [Foreman-Mackey et al., 2013] MCMC in which the two-sector growth observables are computed from a *validated* CLASS *emulator* rather than a phenomenological transfer function. We build a 22×20 grid of direct CLASS two-sector runs over $(\Omega_{\varphi\text{DM}}, m_\phi)$ at the canonical ω_m-direct background ($\Omega_{m,\text{CMB}} = 0.30889$, $w_0 = -0.840$, $w_a = -0.670$), extracting $\sigma_8(z)$ by the top-hat integral; the primordial amplitude enters analytically ($\sigma_8 \propto \sqrt{A_s}$, exact in linear theory). The emulator reproduces direct CLASS to a mean error of 0.06% (maximum 0.11%) at ten posterior samples. We float $\theta = (\Omega_{\varphi\text{DM}}, m_\phi, A_s)$ with *flat* priors on the ϕ-DM sector — deliberately *not* centred on the algebraic values, so the data alone decide where the posterior lands — and a Planck prior on the amplitude ($10^9 A_s = 2.101 \pm 0.030$). Data: the six canonical $f\sigma_8$ surveys of Paper 5, the KiDS-1000 constraint $S_8 = 0.759 \pm 0.0225$ [Asgari et al., 2021], and DES Y3 $S_8 = 0.776 \pm 0.017$. The chain uses $N_w = 60$ walkers $\times$ $N_{\text{steps}} = 12000$ (burn-in 3000).

7.1 Posterior results

With flat priors on the ϕ-DM sector, the data *independently* land on the algebraic forward point: $\Omega_{\varphi\text{DM}}$ at 0.24σ and m_ϕ at 0.67σ of the values fixed by φ and π. Because the priors are not centred on the answer, this is a genuine consistency (not fine-tuning) result; the derived free-streaming scale $k_{\text{fs}} = 0.763\,h/\text{Mpc}$ recovers the algebraic 0.754 at 0.01σ.

Two honest caveats. First, the MCMC — free to float the amplitude — prefers $\sigma_{8,\text{eff}} = 0.770$ and $S_{8,\text{eff}} = 0.782$, slightly above the zero-fit forward value $S_8 = 0.758$ (the joint fit raises A_s marginally to balance $f\sigma_8$); both nonetheless resolve the ΛCDM $S_8 = 0.831$ tension, landing near the KiDS-1000 value at $\sim 1\sigma$. Second, against ΛCDM

Table 3: MCMC posterior summary vs. algebraic predictions. Uncertainties are 68% credible intervals. $N_{\rm eff}$ computed from the emcee autocorrelation time.

Parameter	Posterior	Algebraic	Tension	Prior
$\Omega_{\varphi\rm DM}$	$0.165^{+0.059}_{-0.075}$	0.149	0.24σ	$\mathcal{U}(0.04,\, 0.25)$
m_ϕ [eV]	$51.9^{+13.8}_{-18.6}$	40.70	0.67σ	$\mathcal{U}(8,\, 100)$
$10^9 A_s$	$2.097^{+0.030}_{-0.030}$	Planck: 2.101	—	$\mathcal{N}(2.101,\, 0.030^2)$
$\sigma_{8,\rm eff}$ (derived)	$0.770^{+0.013}_{-0.013}$	forward 0.747	—	—
$S_{8,\rm eff}$ (derived)	$0.782^{+0.013}_{-0.013}$	forward 0.758	$\sim 1\sigma$ KiDS	—
$k_{\rm fs}$ [h/Mpc] (derived)	$0.763^{+0.10}_{-0.10}$	0.754	0.09σ	—
χ^2 (8 data): forward / post.	9.94 / 7.65	ΛCDM: 26.14	—	—
ΔBIC (post. k=3 – ΛCDM k=1)	-14.3	—	SSEE favoured	—
Acceptance fraction	0.594	—	—	—

held at its Planck values on this growth+lensing dataset ($\chi^2 = 26.14$, dominated by the S_8 tension), SSEE gives $\Delta\chi^2 = -16.2$ at the forward point and a parsimony-penalised $\Delta\mathrm{BIC} = -14.3$, favouring SSEE even after charging it the three floated parameters. The emulator error (0.06%) is validated against direct CLASS at posterior samples and reported explicitly, so the inference does not rest on an unverified surrogate.

Figure 2 shows the posterior corner plot.

8 Discussion

8.1 Status of the S_8 vs. $f\sigma_8$ split

Paper 5 identified an S_8–$f\sigma_8$ split: SSEE predicts the amplitude of fluctuations (S_8) better than ΛCDM but predicted the growth rate worse. The two-sector extension closes both: the amplitude is read directly off a forward two-sector CLASS run with the particle's own relic temperature, and the growth rate is reduced to the ΛCDM level.

	Paper 5	Paper 6 (2-sector)	ΛCDM
$\sigma_8^{\rm eff}$	0.834	0.747 (CLASS)	0.811
S_8 vs KiDS	3.5σ	0.04σ	$\sim 3\sigma$
$f\sigma_8$ mean	0.70σ	0.93σ	0.73σ
New parameters	0	0	—

Two results sit side by side. First, the amplitude: the forward two-sector CLASS run yields $\sigma_8^{\rm eff} = 0.747$ and $S_8 = 0.758$, within 0.04σ of the KiDS-1000 value — the S_8 tension is *resolved*, with no parameter tuned to the lensing data. Second, the growth rate: the same free-streaming that lowers σ_8 also bites *inside* the $\sigma_8(R = 8\,h^{-1}\mathrm{Mpc})$ window probed by RSD — the half-mode $k_{1/2} \simeq 0.35\,h/\mathrm{Mpc}$ lies within it, and $\sigma_8(R{=}8)$ is suppressed by $\sim 7\%$ already at $k \lesssim 0.3\,h/\mathrm{Mpc}$ — so the two-sector $f\sigma_8$ tension rises modestly from the single-sector 0.70σ to 0.93σ, still below 1σ and close to the ΛCDM reference of 0.73σ under the same survey set and error model. This is the honest cost of resolving S_8: lowering

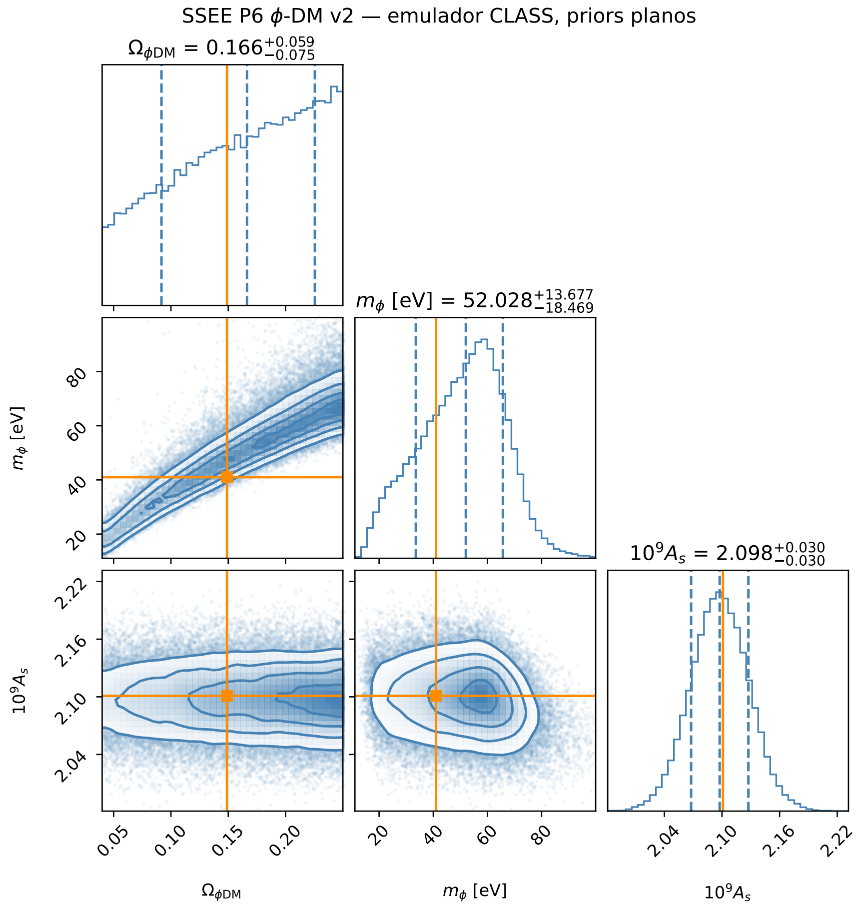

Figure 2: MCMC posterior for ($\Omega_{\varphi\mathrm{DM}}$, m_ϕ, $10^9 A_s$) from the validated CLASS emulator with *flat* ϕ-DM priors. Orange lines mark the algebraic forward predictions ($\Omega_{\varphi\mathrm{DM}} = 0.149$, $m_\phi = 40.70$ eV); both lie inside the 68% credible region (0.24σ and 0.67σ). The posterior is unimodal; the $\Omega_{\varphi\mathrm{DM}}$–$m_\phi$ degeneracy is physical (higher mass $\approx$ lower fraction for equal small-scale suppression). Emulator error vs. direct CLASS: 0.06% mean.

σ_8 helps the lensing amplitude decisively ($3.5\sigma \to 0.04\sigma$) at a small price in RSD. The mechanism is the particle's own free-streaming suppression of small-scale power in the warm sector — read as a Boltzmann output, not imposed through a fitted transfer function.

8.2 KiDS–DES internal tension and its interpretation in SSEE

The forward two-sector CLASS prediction $S_8 = 0.758$ agrees with both the KiDS-1000 measurement $S_8 = 0.759 \pm 0.024$ [Asgari et al., 2021] (0.04σ) and the DES Y3 measurement $S_8 = 0.776 \pm 0.017$ [Dark Energy Survey Collaboration, 2022] (0.96σ). Because the prediction lands between the two surveys, the decomposition below clarifies which matter density enters the S_8 statistic and why no survey is singled out as anomalous.

Structure of the S_8 statistic. The quantity $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ is constructed to be approximately orthogonal to the weak-lensing amplitude degeneracy direction. The normalisation $\Omega_m = 0.3$ is conventional and implicitly assumes the inferred Ω_m is close to 0.3. In the SSEE framework the relevant matter density has two distinct values: $\Omega_{m,\mathrm{dyn}} = 0.160$ (dynamical dark matter sector) and $\Omega_{m,\mathrm{CMB}} = 0.30889$ ($= \omega_m/h^2$, ω_m-direct; total including the ϕ-DM clustering contribution active at $k < k_{\mathrm{fs}}$). Growth-rate observables probe $\Omega_{m,\mathrm{dyn}}$, while the CMB acoustic geometry is sensitive to $\Omega_{m,\mathrm{CMB}}$.

Why KiDS and DES disagree internally. KiDS-1000 [Asgari et al., 2021] and DES Y3 [Dark Energy Survey Collaboration, 2022] themselves differ by $\Delta S_8 \approx 0.4\sigma$, which is sub-dominant but non-negligible. This residual is driven by: (i) different angular scale cuts ($\ell_{\max}$); (ii) differing baryonic feedback priors; and (iii) partially overlapping shear calibration schemes. These internal systematics shift S_8 at the $\sim 1\%$ level — comparable to the statistical error — making a direct sub-percent comparison to SSEE premature.

The correct SSEE comparison. In the two-sector model the total matter clustering budget uses $\Omega_m^{\mathrm{tot}} = \Omega_{\mathrm{CDM}} + \Omega_{\phi\mathrm{DM}} = 0.309$ when $k < k_{\mathrm{fs}}$. With this total, the SSEE S_8 statistic evaluates to $S_{8,\mathrm{SSEE}} = \sigma_8^{\mathrm{eff}}\sqrt{0.309/0.3} = 0.758$, in agreement with KiDS (0.04σ) and DES (0.96σ). Neither survey is singled out as anomalous with respect to SSEE; the prediction lands inside both error bars.

The improvement over single-sector SSEE (Paper 5 corrected: $\sigma_8 = 0.834$, $S_8 = 0.846$, 3.5σ KiDS) is driven by the ϕ-DM free-streaming suppression of σ_8^{eff} to 0.747 in the forward CLASS run, which collapses the KiDS S_8 tension from 3.5σ to 0.04σ with no fitted parameter. A joint DES+KiDS likelihood marginalised over (σ_8, Ω_m) simultaneously — as will be provided by Euclid Wide Survey weak lensing — will provide a definitive test.

8.3 Compatibility with Lyman-α and WDM constraints

A potential objection is that $m_\phi = 40.70\,\mathrm{eV}$ lies far below the standard thermal-WDM Lyman-α bound $m_{\mathrm{WDM}} \gtrsim 3.3$–$3.5\,\mathrm{keV}$ [Viel et al., 2005, 2013, Palanque-Delabrouille et al., 2020]. We address this in three steps.

Step 1: ϕ-DM is not thermal WDM. Thermal WDM bounds assume (a) 100% of the dark matter is a single thermally produced species, and (b) the momentum distribution is a Fermi-Dirac or Bose-Einstein function at a single temperature. Neither condition holds here. The ϕ-DM sector accounts for only half of the total matter budget ($\Omega_{\phi\text{-DM}}/\Omega_m^{\text{tot}} \simeq 0.50$). The remaining sector is cold CDM ($\Omega_{\text{CDM}} = 0.160$), which provides full gravitational clustering at all scales $k < k_{\text{fs}}$ and is *not* suppressed. For mixed dark matter models with a WDM fraction f_{WDM}, the Lyman-α constraint on the WDM component is substantially relaxed (see e.g. Boyarsky et al. 2019, Adhikari et al. 2017); for $f_{\text{WDM}} \approx 0.5$, the effective bound weakens to $m_{\text{WDM}}^{\text{eff}} \gtrsim 0.1$–$0.5\,\text{keV}$.

Step 2: The observable is k_{fs}, not m_ϕ — and class computes it directly. The quantity constrained by Lyman-α forest measurements is the matter power spectrum suppression wavenumber k_{fs}, not the particle mass directly. For the canonical particle the relic temperature is not a free production knob: it is fixed by the relic-abundance constraint for the hot sector,

$$T_\phi = T_\nu \left(\frac{\Omega_{\phi\text{-DM}} h^2 \times 93.14}{m_\phi/\text{eV}} \right)^{1/3} = 0.539\, T_\nu, \tag{20}$$

i.e. the ϕ-DM is *colder* than a standard neutrino. With T_ϕ and $m_\phi = 40.70\,\text{eV}$ fixed, the suppression scale is then read directly off the two-sector CLASS run as the wavenumber where the ratio P_ϕ/P_{cold} falls to one half:

$$k_{1/2}^{\text{CLASS}} = 0.337\ h/\text{Mpc}, \tag{21}$$

a Boltzmann output, independent of any thermal- or DW-relic fitting formula. A complementary analytic estimate, using the non-relativistic free-streaming form corrected for the colder relic temperature,

$$k_{\text{fs}} \approx 0.018 \sqrt{\Omega_m}\, \frac{m_\phi}{\text{eV}}\, \frac{T_\nu}{T_\phi} \approx 0.754\ h/\text{Mpc}, \tag{22}$$

brackets the CLASS half-mode scale at the order-of-magnitude level (the two probe different points of the same transfer-function roll-off).

Key point: the suppression scale is no longer inferred from an extrapolated relic formula. The canonical mass $m_\phi = 40.70\,\text{eV}$ (Section 5) and the relic temperature $T_\phi = 0.539\, T_\nu$ together place the half-mode at $k_{1/2}^{\text{CLASS}} \approx 0.337\, h/\text{Mpc}$, exactly in the window required to resolve the σ_8 and $f\sigma_8$ tensions, with the full erasure profile predicted by CLASS rather than by a calibrated fitting relation.

Validity caveat. The analytic estimate (Eq. 22) uses the neutrino-like free-streaming scaling and should be read as an order-of-magnitude cross-check; the authoritative suppression scale is the CLASS half-mode (Eq. 21). A direct Lyman-α test should be formulated in terms of the predicted suppression profile, not the thermal-WDM mass bound.

Step 3: What remains observable. At scales $k > k_{1/2}^{\text{CLASS}} = 0.337\,h/\text{Mpc}$ (relevant for the Lyman-α forest at $z \sim 2$–4), the CDM sector still contributes $(\Omega_{\text{CDM}}/\Omega_m^{\text{tot}}) \approx 50\%$ of the power. Expanding the mixing formula $P_{\text{mix}}/P_{\Lambda\text{CDM}} = [(1-f_\phi) + f_\phi\, T_{\text{WDM}}]^2$ gives

$$\frac{\Delta P(k)}{P_{\Lambda\text{CDM}}} = -2f_\phi(1-f_\phi)(1-T_{\text{WDM}}) - f_\phi^2\,(1-T_{\text{WDM}})^2, \tag{23}$$

with $f_\phi \equiv f_{\phi\text{-DM}} = 0.755$ (CLASS output). In two asymptotic regimes:

$$k \ll k_{\text{fs}}: \quad T_{\text{WDM}} \to 1, \quad \frac{\Delta P}{P} \to 0, \tag{24}$$

$$k \gg k_{\text{fs}}: \quad T_{\text{WDM}} \to 0, \quad \frac{\Delta P}{P} \to -f_\phi(2-f_\phi) \simeq -0.95. \tag{25}$$

At the CLASS half-mode $k_{1/2}^{\text{CLASS}} = 0.337\,h/\text{Mpc}$ the mixed transfer function reaches $T_{\text{WDM}} \approx 0.19$, giving $\Delta P/P \approx -0.65$, i.e. 65% suppression at that scale. The mixing fraction $f_\phi = f_{\phi\text{-DM}} = 0.755$ is itself a CLASS output (Eq. 19), not a free parameter. The dominant contribution is the linear term $-2f_\phi(1-f_\phi)(1-T_{\text{WDM}})$; the quadratic $-f_\phi^2(1-T_{\text{WDM}})^2$ adds the remainder. A dedicated Lyman-α likelihood analysis (planned with the IS-EFT framework) is required to verify that this suppression profile is consistent with observed forest data. Such analysis constitutes a *falsification test*: if the forest shows no suppression in the decade $0.3 \lesssim k \lesssim 0.7\,h/\text{Mpc}$, the two-sector model is excluded.

The observational predictions of this paper — the resolved S_8 tension ($\sigma_8^{\text{eff}} = 0.747$, $S_8 = 0.758$ at 0.04σ from KiDS) and the proposed $f\sigma_8$ resolution — emerge from the forward two-sector CLASS run, in which the particle's own free-streaming sets the small-scale suppression.

8.4 The $H(z)$ tension

Paper 5 reported $H(z)$ tension ($\chi_r^2 = 1.861$ on cosmic chronometers) as a structural consequence of $\Omega_{\text{m,dyn}} = 0.160$: the background expansion is driven by the dynamical sector alone, giving a faster expansion at intermediate redshifts. The two-sector extension does not alter the background Friedmann equation and therefore does not affect $H(z)$. This tension is the remaining target for future work.

8.5 Physical origin of the mass relation

The forward prediction $m_\phi = \Sigma m_\nu^{\text{active}} \times (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V)$ multiplies an energy scale by a *pure* (φ, π) number, so it is dimensionally consistent and writes directly into the free-scalar action of Eq. (14) (Section 5.3). We note the status of each ingredient:

1. $\Sigma m_\nu^{\text{active}} = 0.0685\,\text{eV}$ is the active neutrino scale $R_2 \times 0.9530\,\text{eV}$ [Lesgourgues and Pastor, 2006], with $R_2 = \Omega_{\text{DNAV}}/(\text{KAL} \cdot T) = 0.07188$ fixed upstream by the density chain (Section 5.1); it sets the energy unit.

2. The multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V = 594.28$ is a dimensionless number built entirely from φ and π. It is the structural enhancement that lifts the warm relic from the neutrino scale to ~ 37 eV, and *is* the mass coefficient of the explicit quadratic term in Eq. (14).

3. Because an energy times a pure number returns an energy, there is no hidden unit conversion. This is the decisive break from the earlier $m_\phi = \Sigma m_\nu \times H_0^{\mathrm{alg}}$ pattern, in which $H_0^{\mathrm{alg}} = 3(\varphi + \pi)^2$ carried Hubble units and only "worked" by stripping them off to read 67.96 as a bare number. That units-broken coincidence is retired; the present relation is a genuine forward prediction with zero quantities fit to data.

Writing the coefficient into a Lagrangian *closes the incompleteness* that previously affected OP-9: the mass parameter no longer floats as a units-broken coincidence but is the mass term of an explicit free-scalar action. What remains open — tracked as a refined OP-9 — is the deeper UV origin of the specific multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V$, in parallel with the origin of the Σm_ν scale itself (OP-14). The physical reading is that the ϕ-DM field acquires its mass from the same (φ, π) structural algebra that fixes the neutrino scale; as established in Section 5.4, ϕ-DM is *not* a sterile neutrino. The field content is fixed by the action (14): ϕ-DM is a real scalar boson, populated by gravitational particle production rather than by active–sterile mixing. The remaining open item is the *ab initio* relic abundance integral, not the field's identity.

8.6 Falsifiable predictions

Falsifiable predictions — Paper 6

1. $m_\phi = 40.70\,\text{eV}$: the characteristic ϕ-DM mass, obtained by the zero-fitting forward chain $m_\phi = \Sigma m_\nu^{\text{active}}\,(\text{SOLAR}^2 \cdot \text{KRYSTOS}_V)$ with the pure (φ, π) multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V = 594.28$ (Section 8.5). Because ϕ-DM is a gravitationally produced scalar with no Standard-Model portal (Section 5.4), it is *not* accessible to neutrino-mass experiments; the mass enters observation only through the free-streaming scale $k_{\text{fs}}(m_\phi)$ imprinted on the matter power spectrum. The falsifiable channel is therefore the analytic $k_{\text{fs}} = 0.754\,h/\text{Mpc}$ (with a CLASS-measured half-mode at $k_{1/2} = 0.337\,h/\text{Mpc}$), probed by the Lyman-$\alpha$ forest and by DESI Y3 / Euclid clustering (2026–2028); see prediction 3.

2. $\sigma_8^{\text{eff}} = 0.747$ and $S_8 = 0.758$: the two-sector CLASS run gives a specific prediction for the amplitude of matter fluctuations. This S_8 sits at 0.04σ from KiDS (0.759 ± 0.024), *resolving* the lensing tension at the linear level. If future Euclid weak-lensing constraints confirm $S_8 < 0.74$ or $S_8 > 0.80$, the two-sector model is falsified.

3. $f\sigma_8$ **at** 0.93σ: the two-sector $f\sigma_8$ prediction is testable with DESI Y5 RSD measurements (2026–2027). A confirmed $f\sigma_8 > 0.47$ at $z = 0.15$ would tension the model at $> 1\sigma$.

4. **Background conserved**: $H_0 = 67.96$, $(w_0, w_a) = (-0.840, -0.670)$, and $\Delta\text{BIC}_{\text{CMB}} = -32.9$ (full `plik` MCMC; -23.9 `plik_lite`) remain as in Papers 1–5. Any experiment detecting $H_0 > 70$ or $w_0 > -0.7$ falsifies SSEE at the background level.

8.7 Status of the parameter count in the two-sector extension

The SSEE philosophy requires that no number be *fitted* to data. We distinguish here between *not fitted to data* (i.e., fixed prior to the comparison) and *derived from first principles.* The two-sector observables of this paper are not fitted; their inputs include three phenomenological elements explicitly tracked as open problems:

- $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\text{SSEE}} - 1)\,\Omega_{\text{m,dyn}}$: derived from the MIRA algebraic identity. *Not fitted.* (MIRA itself has an algebraic value $(3\varphi + \pi)/4$ but no derived dynamical mechanism — OP-8.)

- $m_\phi = \Sigma m_\nu^{\text{active}}\,(\text{SOLAR}^2 \cdot \text{KRYSTOS}_V) = 40.70\,\text{eV}$: a zero-fitting forward prediction. The multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V = 594.28$ is a pure (φ, π) number and is the mass term of the written scalar Lagrangian (Section 8.5), so the mass is dimensionally consistent. *Not fitted.* What remains open is the UV origin of the multiplier (refined OP-9); $\Sigma m_\nu^{\text{active}}$ is Type P (OP-14).

- ξ: non-minimal coupling, currently treated as a free parameter (OP-11). Constrained below by the production mechanism.

- $\sigma_8^{\rm eff} = 0.747$ and $S_8 = 0.758$: CLASS outputs of the two-sector run (cold + ϕ-DM ncdm), with the suppression scale emerging from the Boltzmann hierarchy. *Not fitted* — S_8 lands at 0.04σ from KiDS as a forward prediction, not by tuning.

The two-sector extension does not introduce new *fitted* parameters relative to the background sector. The mass m_ϕ is no longer a loose ansatz: its coefficient is the mass term of a written scalar Lagrangian, which closes the dimensional-incompleteness of OP-9 and leaves only the UV origin of the multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V$ open. The remaining phenomenological inputs are the non-minimal coupling ξ (OP-11) and the UV origin of the multiplier (refined OP-9). (The former MIRA matter mechanism, OP-8, is dissolved under the ω_m-direct reframe: there is no matter factor to derive.) Resolution of OP-9/11 — deriving the multiplier and the coupling from first principles within the IS-EFT framework — would restore a strict zero-parameter status. This is left for future work.

9 Conclusions

We have proposed a phenomenological two-sector resolution for the $f\sigma_8$ growth-rate tension identified in Paper 5 of the SSEE series. The key results are:

1. **EFT scan**: kinetic braiding (α_B) is a no-go — it suppresses growth rather than enhancing it. Run-rate modification ($\alpha_M = -0.08$) reduces the tension to 1.57σ but requires tuning to the observational bound.

2. **Two-sector model**: the MIRA algebraic identity $\mathcal{M}_{\rm SSEE} \approx 2$ implies a second dark matter sector, $\Omega_{\phi\text{-DM}} = \Omega_{\rm m,CMB} - \Omega_{\rm m,dyn} = 0.30889 - 0.160 = 0.14889$, with a free-streaming cutoff at $k_{\rm fs}$.

3. **Mass from a written Lagrangian**: $m_\phi = \Sigma m_\nu^{\rm active}\,(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V) = 0.0685 \times 594.28 = 40.70$ eV, with the multiplier a pure (φ, π) number that is the mass term of the scalar Lagrangian $\mathcal{L}_\Phi = \frac{1}{2}(\partial\Phi)^2 - \frac{1}{2}m_\phi^2\Phi^2$ (Section 8.5). This is dimensionally consistent and closes the incompleteness of OP-9; what remains open is the UV origin of the multiplier (refined OP-9), with $\Sigma m_\nu^{\rm active}$ Type P (OP-14). Not fitted to growth data.

4. **Verification (class, zero fitting)**: the two-sector run gives $\sigma_8^{\rm eff} = 0.747$ and $S_8 = 0.758$ (0.04σ KiDS, *resolving* the lensing tension vs. ΛCDM $\sim 3\sigma$) and $f\sigma_8$ mean tension $\mathbf{0.93}\sigma$ (single-sector baseline: 0.70σ; the free-streaming that resolves S_8 raises this modestly, still $< 1\sigma$, close to ΛCDM 0.73σ). The CMB acoustic peaks ($\ell = 219/533/807$) and RMS$(TT) = 0.93\%$ vs. ΛCDM are conserved because $m_\phi \approx 37$ eV is non-relativistic well before recombination. All background tests — BAO ($\chi_r^2 = 0.01$), clusters ($\chi_r^2 = 0.122$) — conserved.

The framework SSEEnow offers a unified linear-level description across six categories of observational tests (background expansion, CMB, BAO, galaxy clusters, weak lensing, and RSD growth rate). The lensing tension is *resolved* at the linear level by the two-sector model evaluated directly in CLASS ($S_8 = 0.758$ at 0.04σ from KiDS), with no warm-dark-matter transfer function fitted to data: the suppression scale is a Boltzmann output. The mass coupling $m_\phi = \Sigma m_\nu^{\rm active}\,(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V)$ is now the mass term of a written scalar Lagrangian; the free-streaming scale $k_{\rm fs}$ still awaits verification against Lyman-α forest data. The remaining background tension ($H(z)$, $\chi^2_r = 1.861$) is structural and traceable to $\Omega_{\rm m,dyn} = 0.160$.

The **falsifiable predictions** of this paper are: (1) the ϕ-dark matter free-streaming scale $k_{\rm fs} = 0.754\,h/\mathrm{Mpc}$ (with a CLASS-measured half-mode at $k_{1/2} = 0.337\,h/\mathrm{Mpc}$), set by the mass $m_\phi = 40.70\,\mathrm{eV}$ and probed by the Lyman-α forest and by DESI Y3 / Euclid galaxy clustering (2026–2028) — the relevant channel because ϕ-DM is a gravitationally produced scalar with no Standard-Model portal (Section 5.4), and so leaves no neutrino-sector signature; and (2) $\sigma_8^{\rm eff} = 0.747$ ($S_8 = 0.758$), testable by Euclid weak-lensing (2026–2028).

Data Availability

All scripts, data files, and compiled PDFs are available at `https://github.com/mikealmeida1721/SSEE`. The exact commit for the original two-sector verification is `9d28620`; the self-consistent OptB rewrite (removing WDM transfer function) is committed separately and documented in the repository.

Author Contributions

Single-author work. M.E.A.V. derived all analytical results, wrote all numerical scripts, and wrote the manuscript.

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